# 9B 1 REVERSIBLE REACTIONS AND DYNAMIC EQUILIBRIUM

SPECIFICATION REFERENCE 9.9

## LEARNING OBJECTIVES

- Know that many reactions are readily reversible.
- Know that reversible reactions can reach a state of dynamic equilibrium in which:
  - (i) both forward and backward reactions are still occurring
  - (ii) the rate of the forward reaction is equal to the rate of the backward reaction
  - (iii) the concentrations of reactants and products remain constant.

## IRREVERSIBLE AND REVERSIBLE REACTIONS

When a mixture of hydrogen and oxygen in a 2:1 molar ratio is ignited, water is produced. There is very little, if any, uncombined hydrogen or oxygen remaining at the end of the reaction. We often describe such reactions as 'irreversible'.

Most combustion reactions are in this category since they are highly exothermic. That is, $\Delta H$ is large and negative. However, $\Delta H$ is small for many reactions, especially in organic chemistry. These reactions may not go to completion. At the end of the reaction, detectable amounts of the reactants remain, mixed with the product. These reactions are called 'reversible' reactions.

## HOW TO DECIDE WHETHER A REACTION IS REVERSIBLE

Deciding whether a reaction is reversible or not depends on how carefully we measure the concentrations of reactants and products. For example, the reaction between dilute hydrochloric acid and aqueous sodium hydroxide appears to go to completion. Both acid and alkali are almost completely ionised in water, so the equation for the reaction is:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

Pure water has a slight electrical conductivity. This results from the ionisation of water molecules:

$$H_2O(l) \rightarrow H^+(aq) + OH^-(aq)$$

This indicates that the reverse reaction is taking place to a small extent. Since only one molecule in approximately 550 million is ionised, we usually ignore this small extent of ionisation. However, it becomes important when we study the pH scale of acidity.

In practical terms, if a reaction is more than 99% complete, we usually consider it to have gone to completion.

## THE REACTION BETWEEN HYDROGEN AND IODINE

If a mixture of hydrogen and iodine vapour in a 1:1 molar ratio is heated to 573 K in a closed container, about 90% of the hydrogen and iodine react to form hydrogen iodide. Provided the reaction mixture remains in the closed container at 573 K, 10% of the hydrogen and iodine will remain unreacted even if you leave the reaction mixture for a long time.

If a sample of hydrogen iodide is heated to 573 K in a closed container, it partially decomposes. The mixture formed is identical to that produced when starting with an equimolar mixture of hydrogen and iodine. The reaction is clearly reversible, and when there is no further change in the concentrations of the reactants and products, the system is said to be in 'equilibrium'.

The symbol $\rightleftharpoons$ is used in an equation to represent a reversible reaction. The equation for the reaction between hydrogen and iodine is therefore written as follows:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

When the equation is written in this way, the reaction between hydrogen and iodine is called the *forward* reaction. The decomposition of hydrogen iodide into hydrogen and iodine is called the *backward* reaction.

## HOW IS EQUILIBRIUM ESTABLISHED?

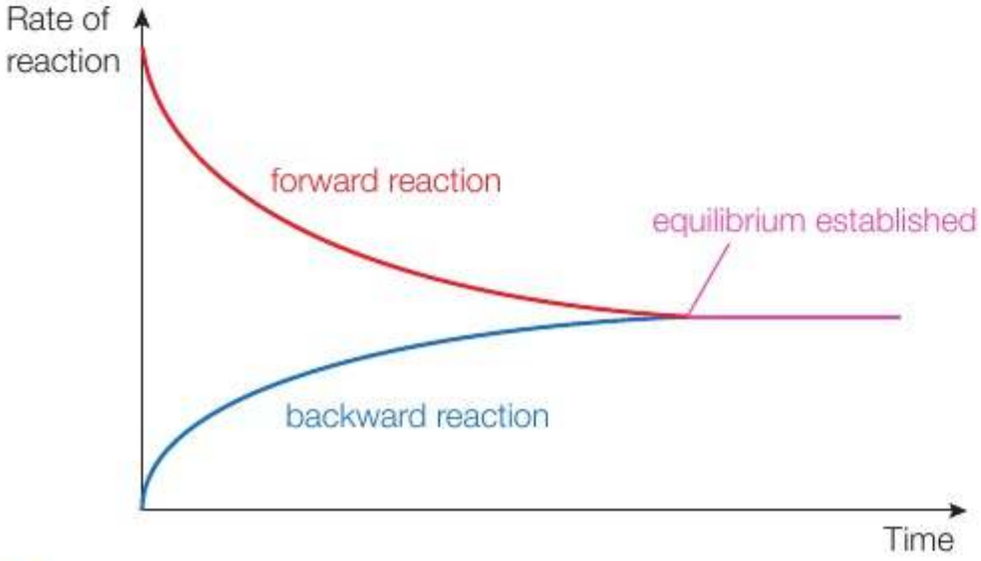
When the mixture of hydrogen and iodine is heated, the two gases start to react and form hydrogen iodide. With increasing time, the concentrations of hydrogen and iodine decrease, so the rate of the forward reaction decreases.

As soon as some hydrogen iodide is formed, it starts to slowly decompose. With increasing time, however, the concentration of hydrogen iodide increases, so the rate of the backward reaction increases.

Eventually, the rates of the forward and the backward reactions become equal. After this point there is no further change in concentrations of reactants and products. The system is now in equilibrium. It is referred to as a 'dynamic' equilibrium since both forward and backward reactions are taking place at the same time, and also at the same rate.

# DYNAMIC EQUILIBRIUM

**Fig A** shows dynamic equilibrium.




▲ **fig A** Dynamic equilibrium.

Two conditions must be met for dynamic equilibrium to be established:

1 the reaction must be reversible

2 the reaction mixture must be in a closed container.

Three important features define a system that is in dynamic equilibrium:

- Both forward and backward reactions are continuously occurring.
- The rate of the forward reaction is equal to the rate of the backward reaction.
- The concentrations of reactants and products remain constant.

**LEARNING TIP**

Remember these features!

## CHECKPOINT

**1.** Explain what is meant by the term 'dynamic equilibrium' with reference to the following reaction:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

**2.** Nitrogen and hydrogen react reversibly to form ammonia:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

1 mol of nitrogen and 3 mol of hydrogen were mixed in a closed container and allowed to reach equilibrium. Twenty per cent of the nitrogen and hydrogen were converted into ammonia.

Draw three graphs, using a single set of axes, to show how the number of moles of nitrogen, hydrogen and ammonia vary with time.

# 9B 2 EFFECT OF CHANGES IN CONDITIONS ON EQUILIBRIUM COMPOSITION

SPECIFICATION REFERENCE 9.10

## LEARNING OBJECTIVES

- Predict and justify the qualitative effect of a change in concentration, pressure or temperature, or the addition of a catalyst, on the composition of an equilibrium mixture.

## CHANGING THE COMPOSITION OF AN EQUILIBRIUM MIXTURE

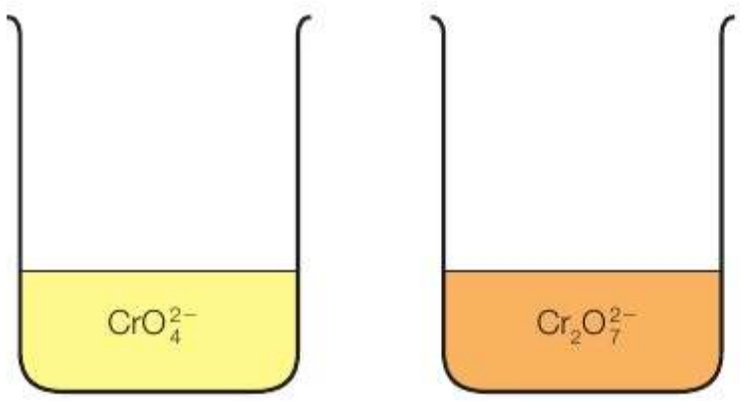
When a reaction mixture reaches a position of equilibrium, the composition of the equilibrium mixture (i.e. the concentration of each component) will not alter as long as the conditions remain the same.

However, if we change a condition (i.e. add some more of, or remove one of, the components, or change the temperature of the system) then the composition may change. This is often referred to as 'changing the position of equilibrium', and we refer to the position being moved to the right, to the left or not changed.

For example, if acid is added to a yellow solution containing chromate(VI) ions, $CrO_4^{2-}$, the solution turns orange. This is because of an increase in the amount of dichromate(VI) ions, $Cr_2O_7^{2-}$.

**EXAM HINT**

Note that there is no change in the oxidation number of the chromium in this reaction.




$$2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$$

yellow orange

▲ **fig A** Beakers containing solutions of chromate(VI) ions and dichromate(VI) ions.

The equilibrium position moves to the right when the acid ($H^+$) is added. If sufficient alkali is added to the orange solution, it will turn yellow as the amount of $CrO_4^{2-}$ ions increases and exceeds the amount of $Cr_2O_7^{2-}$ ions. The equilibrium position moves to the left when alkali ($OH^-$) is added.

We will consider four factors that may affect the position of equilibrium of a reaction mixture. These are:

1 concentration of a component
2 pressure of the system
3 temperature of the system
4 addition of a catalyst.

## EFFECT OF A CHANGE IN CONCENTRATION

If we *increase* the concentration of one of the reactants in a system in equilibrium, the rate of the forward reaction will increase and more products will form. As the concentration of the products increases, the rate of the backward reaction increases and eventually a new equilibrium is established. The equilibrium position has moved to the right, with slightly more product being present than at the original position of equilibrium.

If the concentration of one of the reactants is *decreased*, the position of equilibrium moves to the left. Similar changes occur if the concentration of the product is increased or decreased.

The changes that occur are summarised in the **table A**.

| CONCENTRATION OF REACTANTS | CONCENTRATION OF PRODUCTS | CHANGE IN POSITION OF EQUILIBRIUM |
|---|---|---|
| increased | | to the right |
| decreased | | to the left |
| | increased | to the left |
| | decreased | to the right |

**table A** The change in the position of equilibrium when the concentration of reactants or products is increased or decreased.

## EFFECT OF A CHANGE IN PRESSURE

The effect of pressure only applies to reversible reactions involving gases. At a given temperature, the pressure of a gaseous mixture depends only on the number of gas molecules in a given volume. So, the pressure of a gaseous mixture may be increased by reducing the volume and reduced by increasing the volume.

Alternatively, the pressure at which the reaction is carried out can be:

- increased by initially using more moles of the reactants in the same volume
- decreased by using fewer moles of the reactants in the same volume.

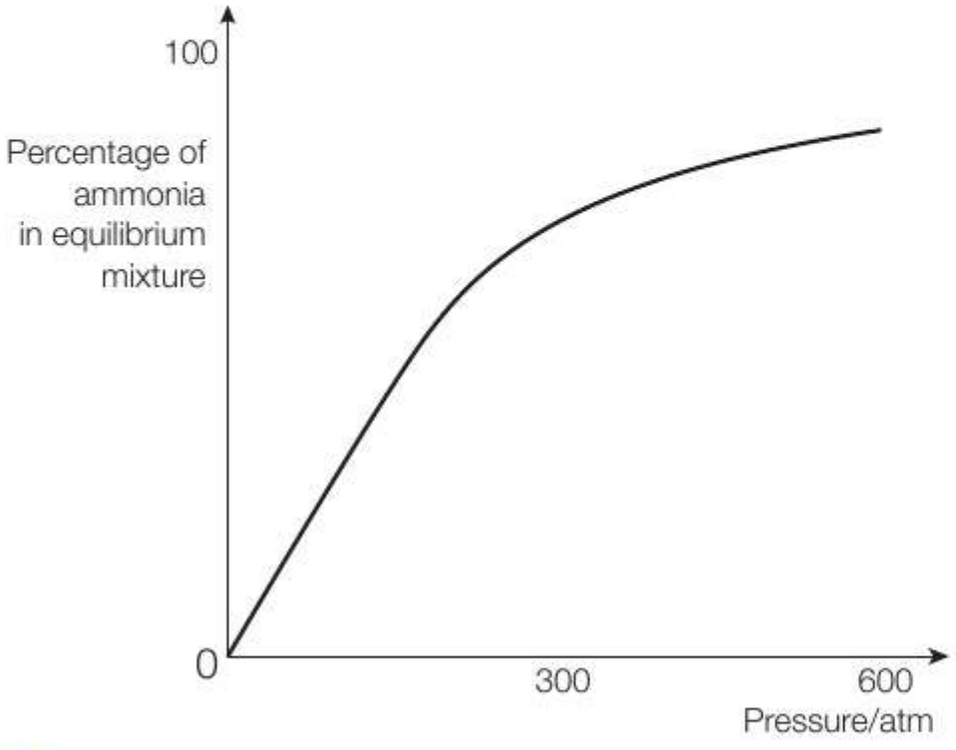
The effect of a change in pressure (at constant temperature) caused by changing the volume of the reaction mixture can be studied using a gas syringe and pushing in, or pulling out, the plunger. The effect depends on the total number of moles of gas on each side of the balanced equation, and is summarised in **table B**.

| NUMBER OF MOLES OF REACTANTS | NUMBER OF MOLES OF PRODUCTS | CHANGE IN POSITION OF EQUILIBRIUM WHEN THE PRESSURE IS INCREASED |
|---|---|---|
| more | fewer | to the right |
| fewer | more | to the left |
| same | same | no change |

**table B** The change in the position of equilibrium when the pressure is increased and the number of moles of reactants or products is increased, decreased or not changed.

The reverse changes are true for a decrease in pressure.

The effect of changes in pressure as a result of using different amounts of gaseous reactants in a fixed volume container is shown in **fig B**. The reaction is between nitrogen and hydrogen, forming ammonia.




▲ **fig B** The effect of pressure changes using different amounts of nitrogen and hydrogen to form ammonia.

The equation for the reaction is:

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

In the balanced equation, there are 4 moles of gas on the left-hand side and 2 moles of gas on the right-hand side.

The graph in **fig B** shows that the higher the pressure, the more ammonia there is in the equilibrium mixture formed. This agrees with the prediction in **table B** which states that an increase in pressure shifts the position of equilibrium to the side that has fewer moles of gas.

**EXAM HINT**

When you answer questions dealing with the effect of temperature change, avoid phrases such as 'an increase in temperature favours the endothermic direction'.
An increase in temperature 'favours' (i.e. increases the rate of) *both* the endothermic *and* the exothermic reactions.

The important point to remember is that it increases the rate of the endothermic reaction *more than* it increases the rate of the exothermic reaction.

## EFFECT OF A CHANGE IN TEMPERATURE

If the temperature of an equilibrium mixture is raised, the rates of both the forward and the backward reactions will increase. However, the increase in the rate of the endothermic reaction will be greater than the increase in the rate of the exothermic reaction. Therefore, an increase in temperature will shift the position of equilibrium in the direction of the endothermic reaction.

So, the change in position of equilibrium will depend on whether the forward reaction is exothermic or endothermic.

The effects of temperature change are summarised in **table C**.

| TEMPERATURE CHANGE | THERMICITY OF FORWARD REACTION | CHANGE IN POSITION OF EQUILIBRIUM |
|---|---|---|
| increased | exothermic ($\Delta H$ –ve) | to the left |
| decreased | exothermic ($\Delta H$ –ve) | to the right |
| increased | endothermic ($\Delta H$ +ve) | to the right |
| decreased | endothermic ($\Delta H$ +ve) | to the left |

**table C** The change in the position of equilibrium and the thermicity of the forward reaction when the temperature changes.

## EFFECT OF THE ADDITION OF A CATALYST

If a catalyst is added to a reaction mixture that is in equilibrium, the rate of both the forward and the backward reactions will increase. However, unlike the effect of increasing the temperature, the increase in rate will be the same for both reactions. So, the position of equilibrium is not altered.

The advantage of adding a catalyst at the beginning of the reaction is that it will reduce the time required to establish equilibrium.

# LIMITATIONS OF MAKING QUALITATIVE PREDICTIONS

The first thing to recognise is that the qualitative predictions we have made about the effect of concentration, pressure and temperature on the position of equilibrium are just that – predictions.

The arguments we have used are *not* explanations of why changes sometimes occur. In fact, there are occasions when it is impossible to predict the direction of change, or indeed when the prediction turns out to be incorrect.

For example, if an equilibrium mixture of $NO_2(g)$ (brown) and $N_2O_4(g)$ (colourless) in a closed container is placed into a beaker of hot water at room temperature, both the temperature and the pressure of the gaseous mixture will rise.

The equation for the reaction is:

$$2NO_2(g) \rightleftharpoons N_2O_4(g) \qquad \Delta H = -57.2\,\text{kJ mol}^{-1}$$

Because the forward reaction is exothermic, we would predict that an increase in temperature would shift the equilibrium to the left.

However, since there are fewer moles of gas on the right-hand side of the equation, we would predict that an increase in pressure would shift the equilibrium to the right.

We do not know which effect is greater, so we cannot make a prediction of which way the equilibrium will shift. In practice, the mixture becomes darker in colour, so the temperature effect must be greater than the pressure effect.This is because the equilibrium must have shifted to the left to produce more brown $NO_2(g)$.

It might be interesting for you to do further research into the limitations of qualitative predictions. For example, what would you expect to happen if you add more nitrogen at constant pressure and temperature to an equilibrium mixture of nitrogen, hydrogen and ammonia? What would you expect to happen if you added an inert gas at constant volume and temperature to an equilibrium mixture of sulfur dioxide, oxygen and sulfur trioxide? The results may surprise you!

**DID YOU KNOW?**

A principle known as Le Chatelier's Principle is sometimes recommended as a useful way of working out the possible change in position of equilibrium when a condition is altered. There is no requirement to learn or use this principle in this course.

Le Chatelier's Principle does not offer an explanation as to why a position of equilibrium alters. Explanations are only possible through the use of the equilibrium constant, *K*. This concept will be introduced in **Topic 13 (Book 2: IAL)**.

## CHECKPOINT

SKILLS REASONING

**1.** Ethanoic acid and ethanol react reversibly to produce ethyl ethanoate and water:

$$CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$$

Assume $\Delta H = 0\,kJ\,mol^{-1}$

(a) Predict the effect on the position of equilibrium of increasing the temperature. Justify your answer.

(b) Suggest why the reaction mixture is heated when used to prepare ethyl ethanoate.

**2.** When carbon dioxide dissolves in water, it forms a solution containing some carbonic acid:

$$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$$

Carbon dioxide is less soluble in hot water than in cold water. Is $\Delta H$ for the forward reaction negative or positive? Justify your answer.

**3.** A sample of insoluble solid lead(II) chloride, $PbCl_2$, is shaken with some dilute hydrochloric acid and left until an equilibrium mixture containing some undissolved lead(II) chloride is established:

$$PbCl_2(s) + 2Cl^-(aq) \rightleftharpoons PbCl_4^{2-}(aq)$$

State what would be observed if concentrated hydrochloric acid was added to the mixture. Justify your answer.

**4.** In each case, predict whether the equilibrium position shifts to the right or the left, or is unaltered when the pressure is increased at constant temperature.

(a) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(b) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(c) $2O_3(g) \rightleftharpoons 3O_2(g)$

(d) $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$

# 9B 3 REVERSIBLE REACTIONS IN INDUSTRY

SPECIFICATION REFERENCE 9.11

## LEARNING OBJECTIVES

- Evaluate data to explain the necessity, for many industrial processes, of reaching a compromise between the yield and the rate of reaction.

## APPLYING THE PRINCIPLES OF REACTION RATES AND REVERSIBILITY TO INDUSTRIAL PROCESSES

The principles of reaction rates and reversibility play an important role in the design and conditions for many industrial processes. In order to maximise profits, the major problems chemists face are to convert the reactants into the products:

- as quickly as possible
- as completely as possible.

The first problem relates to kinetics (rate of reaction) and the second problem relates to reversibility.

The solution to each of these problems requires a careful choice of reaction conditions. This is easily demonstrated by considering the Haber process for the manufacture of ammonia.

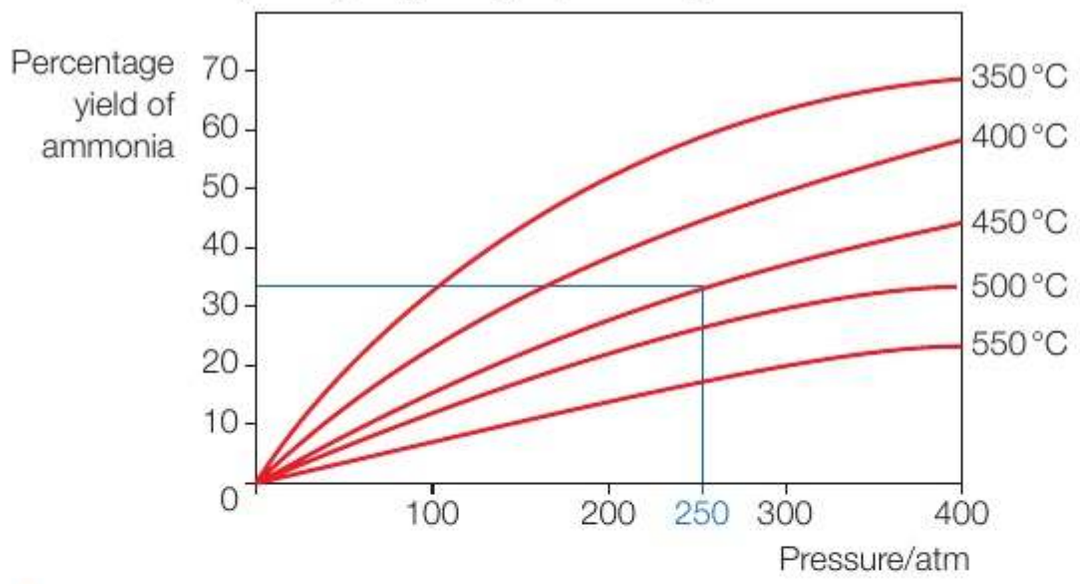
## THE HABER PROCESS

Ammonia is manufactured in industry by direct **synthesis** from nitrogen and hydrogen:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -92\,kJ\,mol^{-1}$$

If the reaction mixture were to reach equilibrium, the maximum yield of ammonia would be obtained by using a low temperature (forward reaction is exothermic) and a high pressure (4 mol of gas on the left; 2 mol of gas on the right).

However, the reaction mixture does not reach a position of equilibrium in the reaction chamber. You can see this by analysing the graphs in **fig A**.




▲ **fig A** Graphs showing the equilibrium yield of ammonia at different temperatures and pressures.

The conditions used in the Haber process are typically 450 °C and 250 atmospheres (atm) pressure. If the reaction mixture were to reach equilibrium, then the yield of ammonia would be just over 30%. In practice, the yield is approximately 15%. This is because the reaction mixture does not remain in the reaction chamber long enough for equilibrium to be established.

## WHY ARE CONDITIONS OF 450 °C AND 250 ATM USED?

The reaction between nitrogen and hydrogen is extremely slow at room temperature. This is mainly because of the very strong nitrogen to nitrogen triple bond [$E(N{\equiv}N) = 945\,kJ\,mol^{-1}$] producing a high activation energy for the reaction. Even at high temperatures, the rate of reaction is slow in the absence of a suitable catalyst. Many metals will catalyse this reaction, including tungsten and platinum. However, these metals are very expensive, so iron is used in the industrial process.

The catalyst does not function very efficiently at low temperatures, so a relatively high temperature is necessary. However, a very high temperature would be uneconomical, because of the extra energy costs involved. It might also result in a decreased yield, since the increase in rate of the backward reaction will be greater than the increase in rate of the forward reaction. This is debatable, because the reaction mixture may not be in the reaction chamber for long enough for this to make a significant difference. For these reasons, a compromise temperature of 450 °C is used.

Under these conditions, the actual yield is about 50% of the equilibrium yield. If we assume this to be true for other pressures at a temperature of 450 °C, then a pressure of 100 atm would give a yield of around 12–13%, while a pressure of 400 atm would give a yield of around 27–28%. The higher the pressure, the larger the energy costs of compressing the gases: the lower the pressure, the lower the yield. Once again, a compromise is reached between yield and cost and a pressure of 250 atm is used.

In order to increase the efficiency of the process, the unreacted nitrogen and hydrogen, after separation from the ammonia, are mixed with fresh nitrogen and hydrogen and fed into the reaction chamber.

## THE CONTACT PROCESS

Sulfuric acid is manufactured by the Contact process. The name of this process comes from the stage of the process that involves the reaction between sulfur dioxide and oxygen at the surface of a solid vanadium(V) oxide, $V_2O_5$, catalyst, to form sulfur trioxide.

$$SO_2(g) + \tfrac{1}{2}O_2(g) \rightleftharpoons SO_3(g) \qquad \Delta H = -96\,kJ\,mol^{-1}$$

The forward reaction is exothermic, so a low temperature would favour a high yield of $SO_3(g)$. You can see this in the graph in **fig B**, which shows the percentage yield of $SO_3(g)$ against temperature at a pressure of 1 atm.

Once again, the catalyst would not be very effective at low temperatures, so a moderately high temperature of 450 °C is used.

At 1 atm pressure, the yield of $SO_3(g)$ is already very high at around 97%. Higher pressures would increase the yield since there are fewer moles of gas on the right-hand side of the equation.

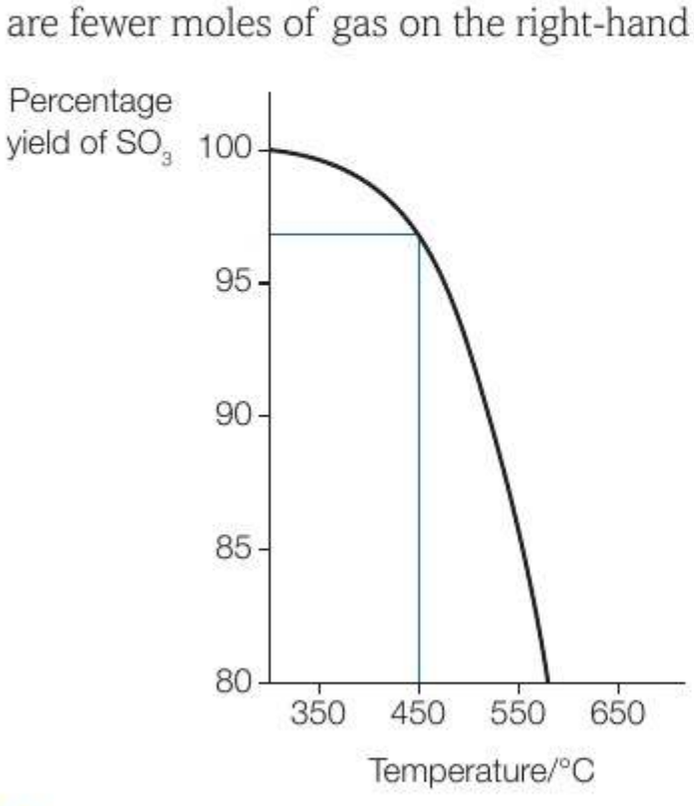



**fig B** A graph showing the percentage yield of sulfur trioxide against temperature at a pressure of 1 atm.

The pressure employed in the Contact process is about 2 atm. This is high enough to maintain a constant flow of gases through the reaction chamber. Pressures higher than this are unnecessary since the yield is already very high.

It is interesting to look at the mechanism of action of the vanadium(V) oxide. Unlike the iron in the Haber process, the vanadium changes its oxidation state during the reaction, but it then converts back to its original oxidation state at the end.

The mechanism for the reaction is:

$$SO_2(g) + V_2O_5(s) \rightleftharpoons SO_3(g) + V_2O_4(s)$$

$$V_2O_4(s) + \tfrac{1}{2}O_2(g) \rightleftharpoons V_2O_5(s)$$

## CHECKPOINT

SKILLS REASONING

**1.** The first step in the manufacture of nitric acid, $HNO_3(l)$, from ammonia involves the exothermic reaction of ammonia with oxygen gas to form nitrogen monoxide NO(g) and steam. This is a reversible reaction that can reach a position of equilibrium.

(a) Write an equation for the reaction between ammonia and oxygen to form nitrogen monoxide and steam.

(b) Assuming the reaction mixture reaches a position of equilibrium, make a qualitative prediction of the conditions of temperature and pressure that would produce the maximum yield of NO(g).

(c) In industry, the conditions used are high temperature and a pressure of around 7 atm. Suggest why these conditions are different to those you predicted in (b) to obtain a high yield of NO(g) in the equilibrium mixture. Again, assume that the reaction reaches a position of equilibrium.

**2.** The densities of diamond and graphite are 3.5 and 2.3 g $cm^{-3}$, respectively.

The change from graphite to diamond can be represented by the equation:

$$C(graphite) \rightleftharpoons C(diamond) \qquad \Delta H = +2\,kJ\,mol^{-1}$$

Suggest the conditions of temperature and pressure that would favour the formation of diamond from graphite. Justify your answers.

## SUBJECT VOCABULARY

**synthesis** the production of chemical compounds by reaction from simpler substances